

Lecture 6/Module 1

DC Circuit Analysis and Theorems

31/07/2026

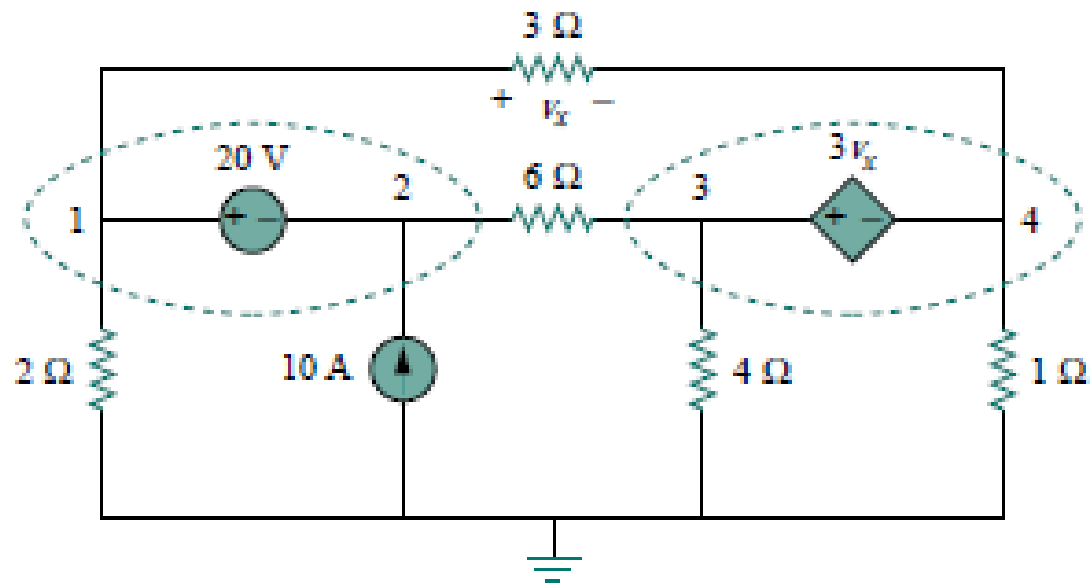
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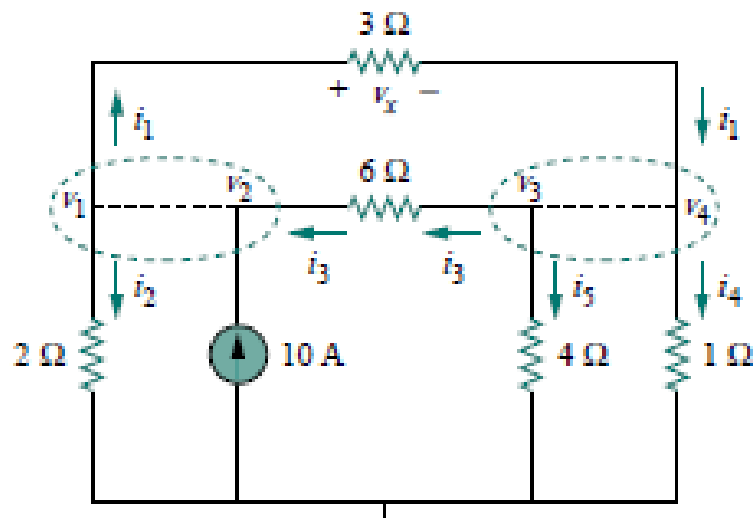


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Find the node voltages in the circuit of Fig. 3.12.



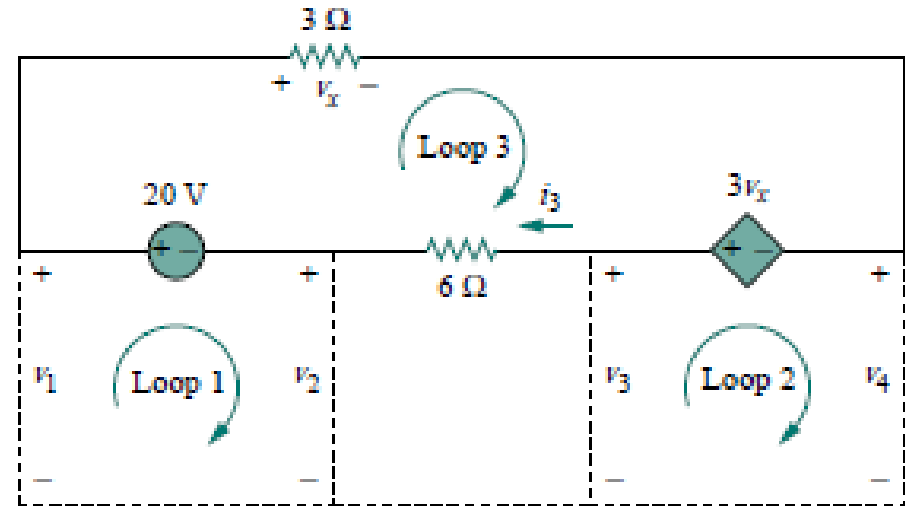
KCL



$$\frac{v_3 - v_2}{6} + 10 = \frac{v_1 - v_4}{3} + \frac{v_1}{2}$$

$$\frac{v_1 - v_4}{3} = \frac{v_3 - v_2}{6} + \frac{v_4}{1} + \frac{v_3}{4}$$

KVL



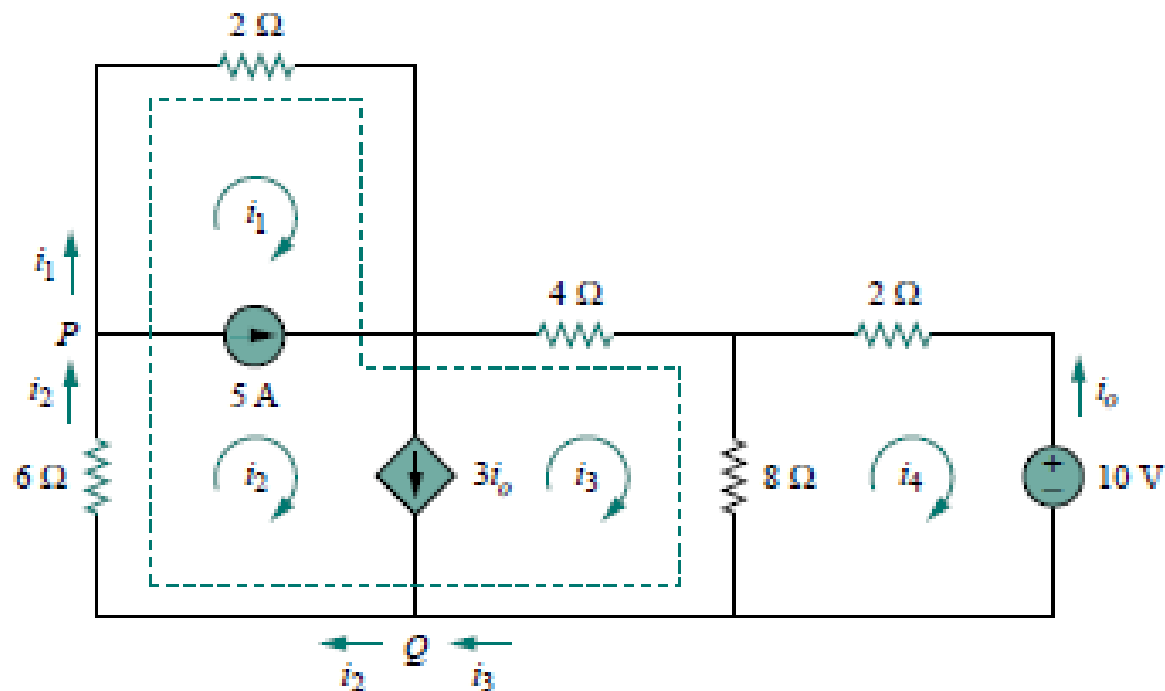
$$-v_1 + 20 + v_2 = 0$$

$$v_x = v_1 - v_4$$

$$3v_1 - v_3 - 2v_4 = 0 \quad 6i_3 = v_3 - v_2$$

$$v_x - 3v_x + 6i_3 - 20 = 0$$

For the circuit in Fig. 3.24, find i_1 to i_4 using mesh analysis.



KVL

$$2i_1 + 4i_3 + 8(i_3 - i_4) + 6i_2 = 0$$

$$2i_4 + 8(i_4 - i_3) + 10 = 0$$

KCL

$$i_2 = i_1 + 5 \qquad i_o = -i_4,$$

$$i_2 = i_3 + 3i_o$$

Superposition Theorem

If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to use **nodal** or **mesh** analysis.

Another way is to **determine the contribution of each independent source to the variable and then add them up**. The latter approach is known as the superposition principle

The Superposition principle

states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

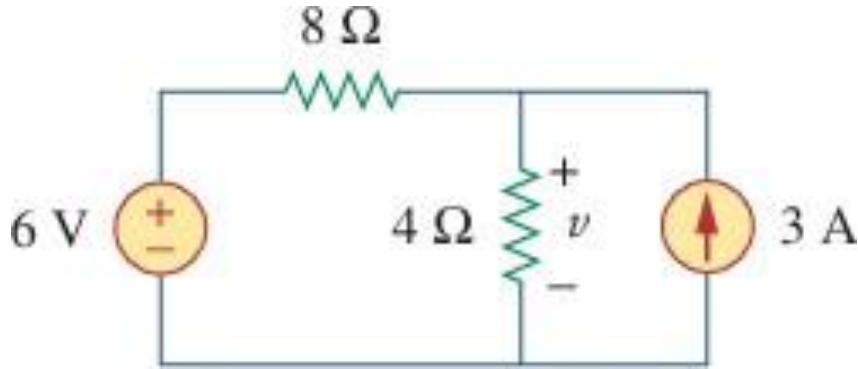
Superposition Theorem

Steps to apply Superposition theorem:

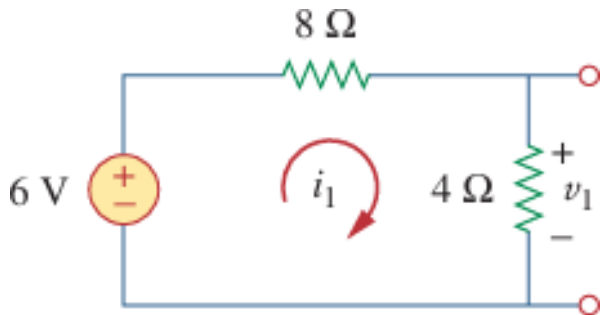
1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using the techniques learnt.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

Superposition Theorem

Find v using superposition theorem.



Let us say, $v = v_1 + v_2$ where v_1 and v_2 are the contributions due to the source 6V and 3A independently.

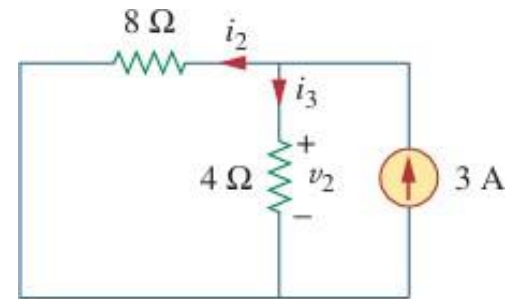


(a)

$$12i_1 - 6 = 0 \Rightarrow i_1 = 0.5 \text{ A}$$

$$v_1 = 4i_1 = 2 \text{ V}$$

Also by using voltage division, $v_1 = \frac{4}{4 + 8}(6) = 2 \text{ V}$



(b)

To get we set the voltage source to zero. By current division,

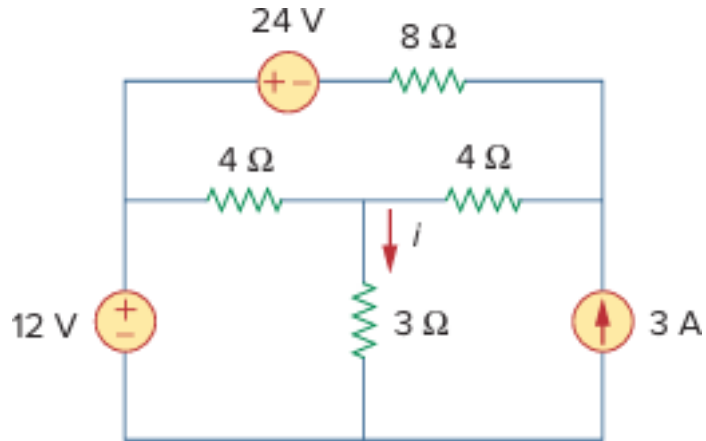
$$i_3 = \frac{8}{4 + 8}(3) = 2 \text{ A}$$

$$v_2 = 4i_3 = 8 \text{ V}$$

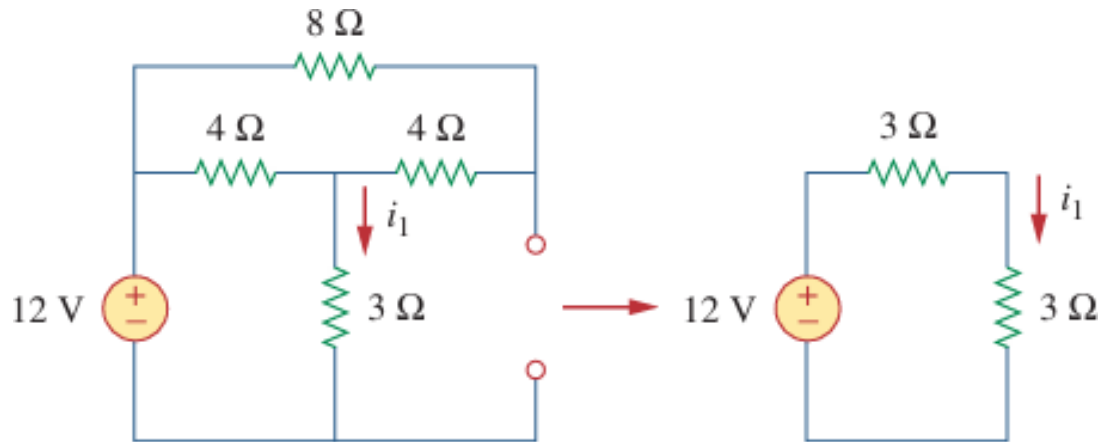
$$\text{Hence, } v = v_1 + v_2 = 2 + 8 = 10 \text{ V}$$

Superposition Theorem

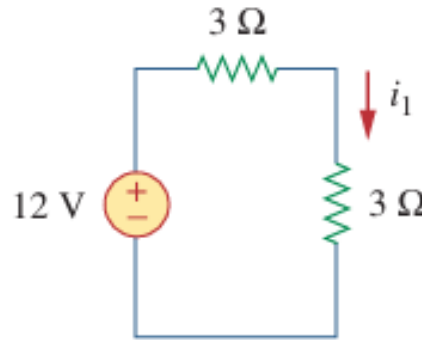
For the circuit, use the superposition principle to find i .



Let us say $i = i_1 + i_2 + i_3$



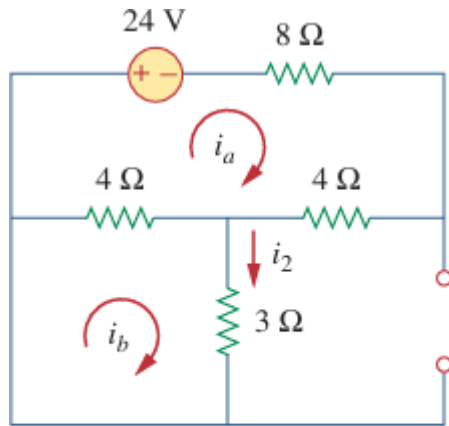
(a)



$$i_1 = \frac{12}{6} = 2 \text{ A}$$

Superposition Theorem

To get i_2 , consider the circuit and apply mesh analysis.



(b)

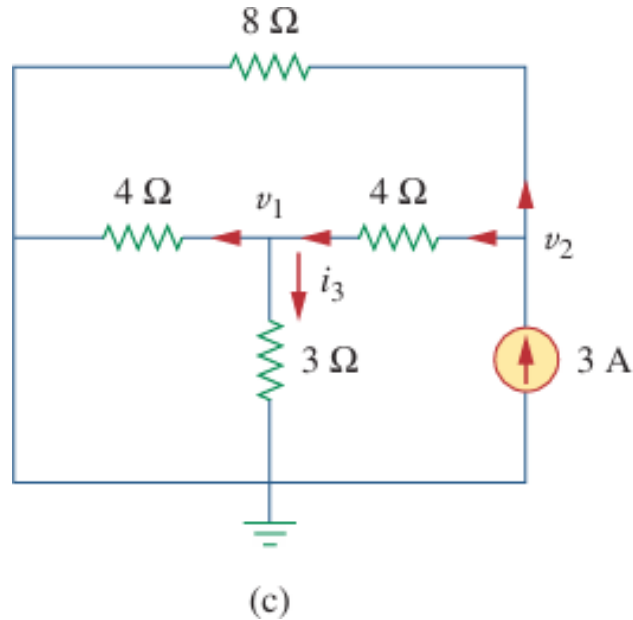
$$16i_a - 4i_b + 24 = 0 \quad \Rightarrow \quad 4i_a - i_b = -6$$

$$7i_b - 4i_a = 0 \quad \Rightarrow \quad i_a = \frac{7}{4}i_b$$

$$i_2 = i_b = -1$$

Superposition Theorem

To get i_3 , consider the circuit and use nodal analysis.



$$3 = \frac{v_2}{8} + \frac{v_2 - v_1}{4} \Rightarrow 24 = 3v_2 - 2v_1$$

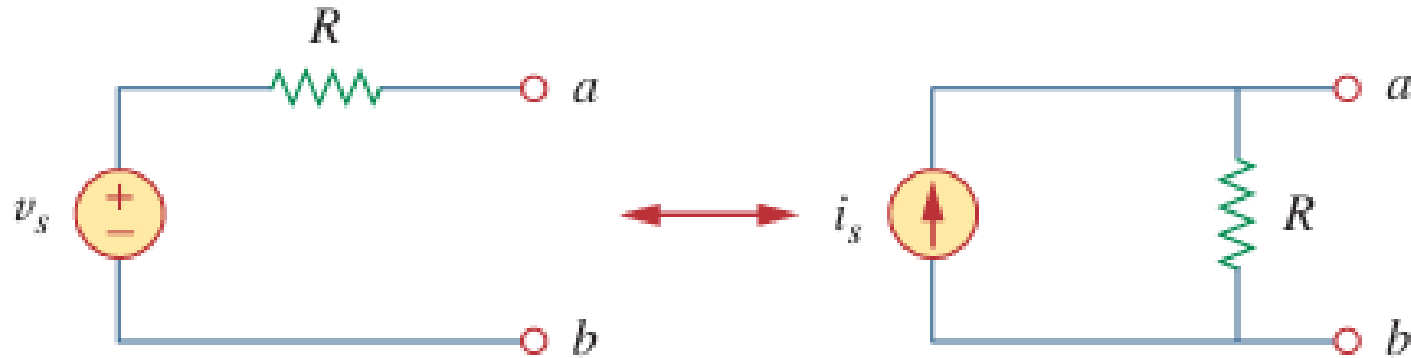
$$\frac{v_2 - v_1}{4} = \frac{v_1}{4} + \frac{v_1}{3} \Rightarrow v_2 = \frac{10}{3}v_1$$

$$v_1 = 3$$

$$i_3 = \frac{v_1}{3} = 1 \text{ A}$$

$$i = i_1 + i_2 + i_3 = 2 - 1 + 1 = 2 \text{ A}$$

Source Transformation



1. Note that the arrow of the current source is directed toward the positive terminal of the voltage source.
2. Note that source transformation is not possible when $R = 0$, which is the case with an ideal voltage source. However, for a practical, non-ideal voltage source, R is not equal to 0 .
3. Similarly, an ideal current source with $R = \infty$ cannot be replaced by a finite voltage source.

Source Transformation

The two circuits are equivalent—provided they have the same voltage-current relation at terminals a-b.

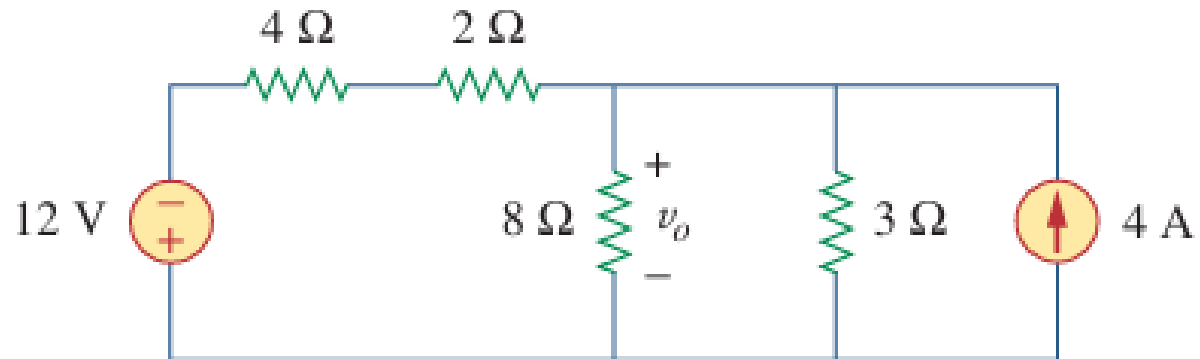
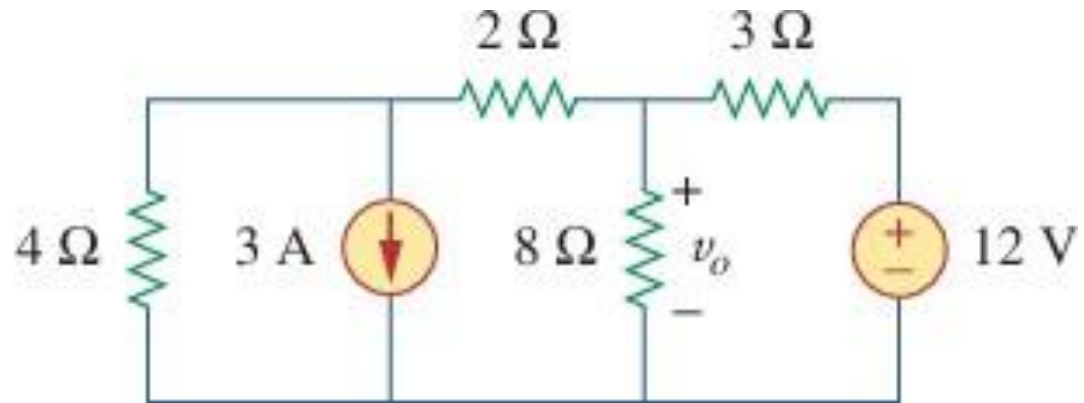
If the sources are turned off, the equivalent resistance at terminals a-b in both circuits is R .

Hence, source transformation requires that,

$$v_s = i_s R \quad \text{or} \quad i_s = \frac{v_s}{R}$$

Source Transformation

Use source transformation to find v_o in the circuit.



Thank You